

Fake Engagement Detection Across Social Media Platforms

Group Project Report

Module	CCS4310 – Deep Learning
Group Members	Tharanya Pushparaj (CIT-23-02-0176) Amintha Jayasooriya (CIT-23-02-0335) Damsara Dissanayaka (CIT-23-02-0163) Thamindu Kavinda (CIT-23-02-0356)
Module Owner	Mr. Indeera Weerasinghe
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01 PROJECT INTRODUCTION

1.1 Project Title

Detecting Fake Engagement Across Social Media Platforms: A Machine Learning and Deep Learning Approach to Identifying Inauthentic Likes, Comments, Followers, and Views

1.2 Project Overview

Each major social networking platform processes user engagement differently. Instagram and TikTok are frequently affected by purchased followers and coordinated engagement pods, while Twitter/X often experiences automated bot networks and coordinated political amplification. Similarly, YouTube and TikTok face challenges with artificially inflated video views generated through automated systems.

This project focuses on automatically detecting fake engagement generated by automated sources rather than genuine human interactions across five major social media platforms:

- Facebook
- Instagram
- TikTok
- Twitter/X
- YouTube

The proposed system combines both classical Machine Learning (ML) and modern Deep Learning (DL) techniques to identify engagement fraud while developing models capable of generalising across multiple platforms.

Classical ML methods include behavioural and statistical feature extraction with classifiers such as:

- Random Forest
- Support Vector Machine (SVM)
- XGBoost

Deep Learning approaches include:

- Long Short-Term Memory (LSTM)
- Gated Recurrent Unit (GRU)

- Transformer models (DistilBERT, RoBERTa)
- Graph Neural Networks (GNN)

These techniques analyse sequential engagement behaviour, detect repetitive bot-generated comments, and identify suspicious follower-network structures.

1.3 Problem Statement

Manual auditing of engagement quality is slow, labour-intensive, and increasingly ineffective against sophisticated fraudulent techniques such as automated bots, engagement pods, comment manipulation, and drip-feeding behaviour. The challenge becomes even greater when monitoring multiple social media platforms with different APIs, data formats, and fraud patterns.

The proposed system aims to:

- Detect whether likes, comments, followers, and views are genuine or fake.
- Classify user accounts as authentic or inauthentic.
- Generate an Engagement Authenticity Score for accounts and posts.
- Explain which behavioural signals contributed to each prediction, including comment text, timing patterns, follower relationships, and network structure.

Research from both industry and academia highlights the growing scale of engagement fraud. Approximately 37% of influencer followers on Instagram and TikTok are estimated to be fake. In 2024, 59.8% of brands reported experiencing influencer fraud compared with 31.0% in 2022. Meta removed approximately 1.4 billion fake Facebook accounts in a single quarter during 2024, while industry reports estimated invalid video traffic to reach nearly 24.55%.

With global influencer marketing expenditure estimated between \$24 billion and \$32 billion during 2024–2025, fraudulent engagement represents billions of dollars in wasted advertising expenditure annually. Therefore, developing an automated cross-platform fake engagement detection framework has significant commercial and academic value.

1.4 ML & DL Application Scope

Machine Learning Components	Deep Learning Components
Feature Engineering: Follower/Following Ratio, Frequency of Posts, Age of Account and the Completeness of Account Bio (Where Possible to be Platform Agnostic)	Model Types: LSTM/GRU for Analyzing Sequential Patterns of Engagement Timing.
Types of Classifiers: Random Forest Classifiers, SVM Classifiers, and XGBoost Classifiers	DistilBERT/RoBERTa for Detecting Bot Comment Text across Multiple Social Media Platforms.
Use of K-Fold Cross Validation Pipeline for Consistent Results Across All Classifiers	Autoencoder Models to Detect Anomalous Bursts of Engagement.
Use of SMOTE for Class Imbalance	Graph Neural Network Models (GNN/RGCN) to Analyze the Structure of the Follower Network Among Different Social Media Platforms (Focus will Be Primarily on Twitter & X, the Most Accessible Network Data Source).
Metrics for Evaluation: Precision, Recall, F1 Score, and AUC-ROC Curve Values	Hybrid Ensemble Machine Learning (ML) plus Deep Learning (DL) Approach to Model Prediction and Transfer Learning across Social Media Platform

1.5 Expected Outcomes

- An analysis comparing ML and DL strategies for detecting fraudulent engagement separately by their function on Instagram, Twitter/X, TikTok/YouTube.
- Using an NLP-based module to identify and flag duplicate, template, or low-quality/comment text that contains identical (or similar) snippets will be tested for transferability across multiple social media platforms.
- Creating a composite engagement algorithm with combined historical, textual, and (in the cases where data is available) network-level metrics.
- Evaluation of the extent that elements which exhibit transferability and those that require a social media-specific tuning process.
- A web-based app prototype used to provide an interface for a user to select their desired social media platform and input handle or URL into a real-time engagement verification process.
- Submission of the peer reviewed documentation for appropriate academic use and citation.

02 LITERATURE REVIEW

The project’s primary literature foundation consists of five key papers published from 2021–2026, one per module requirement, covering developing Instagram and Twitter recognizable cross-platform models because the project is platform neutral. Each paper has been analysed based on methodology, key contributions, limitations, and their relevance to the proposed model.

Paper 1

Reference:

Feng, S., Tan, Z., Wan, H., Wang, N., Chen, Z., et al. (2022). *TwBot-22: Towards Graph-Based Twitter Bot Detection*. NeurIPS 2022 Datasets and Benchmarks Track.

Category	Description
Approach	This paper builds (for the first time in history) the largest available benchmark of graph-based Twitter bot detection (1 million users with 4 entity types, and 14 relation types), along with an extensive two-stage controlled expansion sampling strategy and weakly-supervised annotation (supplied with an additional 1,000 expert labels) that was implemented to remedy both problems of scale (1 million users, 4 entity types, and 14 relation types) and the quality of annotations used in earlier datasets.
Key Contribution	This paper re-implements and benchmarks 35 existing bot detection baselines (feature-based, text-based, graph-based) on a unified evaluation framework that allows for a fair and reproducible comparison of all methods for the first time at this scale.
Limitation	The paper is limited to Twitter, and therefore none of its best results of heterogeneous graph structures (follow, retweet and mention relationships) exist in the same format on Instagram, TikTok, and/or YouTube, which rules out the possibility of transferring its architecture directly to any of those other platforms.
Relevance to Project	This paper provides the primary cross platform benchmark/dataset source for this project’s Twitter element, and the baseline dataset/ML models (e.g. SGBot, BotHunter, etc.) are the models against which our own model will be compared.

Paper 2

Reference:

Feng, S., Wan, H., Wang, N., & Luo, M. (2022). *BotRGCN: Twitter Bot Detection with Relational Graph Convolutional Networks*. Proceedings of ASONAM 2021.

Category	Description
Approach	From using relationships from Twitter, builds heterogeneous user graphs that represent both follow and friendship relationships among users. The use of a Relational Graph Convolutional Network (RGCN) to aggregate neighborhood information differently for each relation type and to combine relational knowledge in the form of multiple relational types.
Key Contribution	Relation-aware graph convolution outperforms any relation-agnostic GCN baseline for bot detection and achieves a 84.6% accuracies on TwiBot-20. Therefore, the study introduces one of the leading reference architectures for graph-based bot detection systems.
Limitation	In order to build the graphs, follow/friend relationship data must be available, but is becoming limited due to major social networks having more control over their APIs post-2022. As discussed in Paper 4, this continues to be a limitation of research that seeks to perform cross-platform analyses.
Relevance to Project	Our adapted GNN architecture for the Twitter/X portion of our hybrid model and the GNN architecture provides an important baseline for assessing the accuracy of our graph-based results (84.6%).

Paper 3

Reference:

Nevado-Catalán, D., Pastrana, S., Vallina-Rodriguez, N., & Tapiador, J. (2023). *An Analysis of Fake Social Media Engagement Services*. Computers & Security, 124, 103013.

Category	Description
Approach	The authors gathered data on 58 active SMM panels—i.e., underground services selling likes, followers, views, and comments across said social media platforms (Instagram, Facebook, TikTok, YouTube, etc.) over a period of several months to create a longitudinal database of their pricing, service descriptions, and methods of delivery for each of the platforms (e.g., "drip," "no-refill," "HQ/LQ" quality tiers).
Key Contribution	The authors performed the first large scale empirical analysis of the fake engagement supply chain itself, as opposed to merely detecting it, using a single data set to analyze fake engagement across multiple social media platforms and thereby facilitating multi-platform comparisons of fraud patterns.
Limitation	The authors produced a description of the supply side of fraud, but did not create a model for detecting it; therefore, the authors suggest that the panel data may not accurately depict the purchased engagement through informal or emerging channels.
Relevance to Project	Panel data provides valuable information for building cross-platform feature engineering; that is, panel data allows us to compare how purchased engagement appears as "realistic" on each of the three social media platforms, because many of the panels offer their services for Instagram, TikTok and YouTube in a side-by-side manner.

03 AVAILABLE DATA SOURCES

The datasets listed here have been purposefully selected to ensure coverage of multiple platforms (Twitter/X, Instagram, and cross-platform engagement markets). This enables the proposed model to be trained and evaluated in a truly cross-platform setting, aligned with the project’s extended scope.

Dataset 1: TwiBot-22 (Twitter/X Bot Detection Benchmark)

Attribute	Details
Access URL	github.com/LuoUndergradXJTU/TwiBot-22 Paper: arxiv.org/abs/2206.04564
Size	1,000,000 users; 86.7M tweets; 170.2M relationships (4 entity types, 14 relation types)
Format	JSON/graph format with evaluation framework and associated benchmark code
Intended Use	Primary cross-platform anchor dataset for Twitter/X. Used for training and benchmarking GNN-based (RGCN) and hybrid ML/DL bot detection models.

Dataset 2: Instagram Fake Spammer/Genuine Accounts (Kaggle)

Attribute	Details
Access URL	kaggle.com/datasets/free4ever1/instagram-fake-spammer-genuine-accounts
Size	Approximately 696 profiles (pre-split into training and testing sets)
Format	CSV format containing 12 numerical and categorical account-level features
Intended Use	Instagram baseline dataset for account-level classification and cross-platform feature transfer comparison with TwiBot-22.

Dataset 3: SMM Panels Dataset (0xjet/smmpanels)

Attribute	Details
Access URL	github.com/0xjet/smmpanels
Size	58 panels with daily CSV snapshots collected over several months
Format	CSV format containing fields such as Site, Product, Price/1000, Min, Max, and service-quality indicators
Intended Use	Analysis of the fake engagement supply ecosystem across platforms. Enables comparison of fraud pricing and service characteristics across Instagram, TikTok, Facebook, and others.

Dataset 4: Instagram Fake Followers Dataset (Kaggle)

Attribute	Details
Access URL	kaggle.com/datasets/bugrayan/instagram-fake-followers
Size	Several thousand follower-level records
Format	CSV containing follower-level numerical behavioral features
Intended Use	Follower-level fake detection on Instagram. Used as secondary validation dataset complementing account-level datasets.

Dataset 5: Influencer-Audience Engagement Network Dataset

Attribute	Details
Access URL	Referenced in: <i>Detecting Engagement Bots on Social Influencer Marketing (multi-relational network study)</i>
Size	14,221 influencers; 9,290,895 audience accounts; 65,848,717 engagements
Format	Graph-structured dataset containing influencer–audience interactions and comment text
Intended Use	GNN-based experiments and large-scale validation of behavioural and textual bot detection signals across social media platforms.

04 DATASET SUMMARY

Cross-Dataset Comparison

Dataset Name	Platform	Granularity	Size	Primary Use
Twibot-22	Twitter/X	Account / Graph	1M users, 170M relations	GNN-based bot detection benchmark
Instagram Fake/Spammer/Genuine (Kaggle)	Instagram	Account	~696 profiles	Account-level classification baseline
SMM Panels Dataset (GitHub)	Multi-platform	Service / Panel	58 panels, daily snapshots	Cross-platform fraud supply analysis
Instagram Fake Followers (Kaggle)	Instagram	Follower	Thousands of records	Follower-level classification
Influencer-Audience Network	Cross-platform	Network Graph	65.8M engagements	Large-scale GNN-based detection

Column-Level Feature Summary — Primary Dataset

The following table describes the key features of the primary Instagram Fake Spammer/Genuine dataset used for training the account classification model.

Column	Meaning	Type	Range / Notes
profile pic	Whether the account has a profile picture set	Categorical (binary)	0 = no, 1 = yes
nums/length username	Ratio of numeric characters in username	Numerical (continuous)	0.0 – 1.0
fullname words	Number of words in full name	Numerical (discrete)	Integer count
nums/length full-name	Ratio of numeric characters in full name	Numerical (continuous)	0.0 – 1.0
name==username	Whether full name equals username	Categorical (binary)	0 = no, 1 = yes
description length	Length of bio/description	Numerical (discrete)	Character count
external URL	Whether bio contains external link	Categorical (binary)	0 = no, 1 = yes
private	Account privacy setting	Categorical (binary)	0 = public, 1 = private
posts	Number of posts	Numerical (discrete)	Integer count
followers	Follower count	Numerical (discrete)	Integer count
follows	Following count	Numerical (discrete)	Integer count
fake	Target label (fake or genuine)	Categorical (binary)	0 = genuine, 1 = fake

5. Conclusion

The proposed project presents a framework for detecting fraudulent engagement across multiple social media platforms using a combination of machine learning and deep learning techniques. By integrating behavioural signals, textual analysis, and network-based features, the system aims to improve the accuracy of identifying fake followers, likes, comments, and views.

A hybrid ML/DL approach enables cross-platform generalisation while maintaining strong performance on platform-specific patterns. The use of multiple independent datasets ensures robustness and scalability of the model across different environments.

Ultimately, the successful implementation of this system will support researchers, organisations, and social media platforms in detecting and reducing fraudulent activity, thereby improving trust and authenticity in online engagement ecosystems.